
Quantization and Multilingual Safety in Low-Resource Languages: Capability Degradation, Safety Behavior, and Evaluation Artifacts

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Abstract

1 Post-training quantization enables efficient deployment of large language models,
2 but its effects on multilingual safety, particularly in low-resource languages, re-
3 main unclear. We conduct a pre-registered audit of 4-bit NormalFloat (NF4) quan-
4 tization in Qwen3-4B and Llama-3.1-8B-Instruct across 10 languages, evaluating
5 multilingual capability and safety refusal. NF4 substantially degrades Llama-3.1-
6 8B capability, with losses increasing from -9.4 percentage points in English to
7 -13.7 points in low-resource languages, supporting our capability-degradation
8 hypothesis. In contrast, we find no evidence of a low-resource-specific safety-
9 refusal effect ($\beta = 0.027$, $p = 0.88$), and capability loss does not correlate with
10 refusal changes ($r = 0.010$, $p = 0.97$). An apparent -58.9 -point refusal col-
11 lapse in Llama low-resource languages reproduces under a verified NF4 run, but
12 analysis of 945 outputs indicates severe generation degeneration and automated
13 scoring artifacts rather than actionable harmful assistance. Blinded human valida-
14 tion ($n = 200$) and tri-stream sensitivity analysis further support the robustness of
15 these findings.

16 1 Introduction

17 Deploying large language models (LLMs) in resource-constrained environments often requires post-
18 training weight quantization [Dettmers et al., 2023]. Low-resource languages face limited native-
19 language safety coverage [Yong et al., 2023, Li et al., 2024], while the same languages can experi-
20 ence larger capability losses under quantization. We test whether 4-bit NormalFloat (NF4) quan-
21 tization accelerates this disparity in underrepresented languages (evaluated on Bengali, Swahili, and
22 Javanese). We evaluate benchmark-level safety refusal and reading comprehension; our data estab-
23 lish a benchmark-level empirical phenomenon rather than population-level social harms.

24 To distinguish generation pathology from benchmark scoring, we use a **behavioral and**
25 **measurement interpretation:**

26
$$\text{NF4} + \text{LRL} \xrightarrow{\text{degeneration}} \text{Repetition Loops} \xrightarrow{\text{scoring failure}} \text{Scoring Error} \xrightarrow{\text{metric}} \text{Apparent "Non-Refusal"}$$

27 These associations do not establish mediation or show that the model provided harmful assistance;
28 instead, they separate output degeneration from automated scoring.

29 We evaluate two pre-registered hypotheses: **H1 (Safety Degradation)**, testing whether NF4 selec-
30 tively degrades refusal on safety relative to control tasks, amplified in low-resource languages (three-
31 way Precision $\times$ Tier $\times$ Task interaction); and **H2 (Capability Degradation)**, testing whether NF4
32 comprehension loss scales monotonically with linguistic resource distance from English. Our audit
33 yields three primary findings: **(1) Monotonic Capability Loss:** NF4 degrades multilingual capa-
34 bility, scaling monotonically with linguistic distance for Llama-3.1-8B (up to -13.7 pp in LRL;
35 supporting H2). **(2) Absence of Disproportionate Safety Loss:** We find no evidence for the pre-
36 registered low-resource safety-refusal interaction in non-gated data ($\beta = 0.027$, $p = 0.88$); refusal
37 changes do not track capability loss ($r = 0.010$, $p = 0.97$). **(3) Evaluator Artifact Conflation:** An
38 apparent -58.9 pp safety-refusal collapse reproduces under the verified v2 protocol and is shown

to be substantially associated with low-resource output degeneration that distorts automated refusal scoring.

2 Related Work

Quantization in Large Language Models. Weight quantization reduces memory bandwidth bottlenecks by discretizing floating-point tensors into low-bit representations [Lin et al., 2023, Frantar et al., 2022]. NF4 [Dettmers et al., 2023] constructs an optimal quantile distribution for normally distributed weights. Existing evaluations focus almost exclusively on English benchmarks, leaving non-English alignment unexplored.

Cross-Lingual Alignment and Evaluator Fragility. Multilingual jailbreak studies show that frontier LLMs often fulfill harmful requests translated into low-resource languages [Yong et al., 2023, Deng et al., 2023], driven by uneven cross-lingual representations [Li et al., 2024]. Recent work shows that LLM safety evaluators are vulnerable to superficial input artifacts [Chen & Goldfarb-Tarrant, 2025] and exhibit poor reliability in low-resource languages [Fu & Liu, 2025], motivating our investigation of evaluator-degeneration interactions.

Mechanistic Representations of Safety. Refusal in aligned models often operates via low-dimensional directional subspaces (Geva et al., 2023), suggesting a hypothesis for why safety refusal and comprehension might show differential sensitivity to weight compression.

3 Methods and Experimental Architecture

Models and Inference. We evaluate **Qwen3-4B** and **Llama-3.1-8B-Instruct** in native BF16 and 4-bit NF4 via BitsAndBytes using greedy decoding (temp=0, seed=42, max_tokens=2048). Confirmatory Llama-3.1-8B-NF4 inference uses a fresh provenance-verified reproduction arm on an NVIDIA H100 GPU (meta-llama/Llama-3.1-8B-Instruct revision 0e9e39f, BitsAndBytes 0.50.2, PyTorch 2.14, Transformers 5.16.1; llama_nf4_v2), retaining historical runs as replication evidence.

Tasks, Datasets, and Stratification. The **safety task** uses MultiJAIL [Deng et al., 2023] (315 parallel adversarial prompts $\times$ 10 languages = 3,150 items per model variant). The **control task** uses Belebele [Bandarkar et al., 2023] (900 parallel 4-choice items $\times$ 10 languages = 9,000 items per model variant). Languages span four tiers [Joshi et al., 2020]: **Reference** (en), **HRL** (zh, it, vi), **MRL** (ar, ko, th), and **LRL** (bn, sw, jv).

Tri-Stream Refusal Adjudication. Automated classifiers like WildGuard [Han et al., 2024] can conflate non-refusal with format degeneration. We establish targeted blinded human validation ($n=200$ frozen items in human_review_packet_blinded_FINAL.csv) alongside three blinded AI judges (Terra GPT-5.6, Google Gemini, Qwen-14B). Human adjudications serve validation/governance rather than gold-standard ground truth. We evaluate: Stream A (automated baseline); Stream B (human-primary adjudication with Jeffreys-smoothed cell priors); and Stream C (worst-case boundary reclassifying all AMBIGUOUS responses as NON_REFUSAL).

Llama NF4 Governance Gate. Automated scoring of Llama-3.1-8B-NF4 in LRL indicated an apparent -58.9 pp drop. Root-cause audit revealed degenerate token repetition loops scored as non-refusal despite containing zero harmful compliance. This stratum was quarantined behind a **Governance Gate** (PRE_ADJUDICATION_DESCRIPTIVE) and excluded from confirmatory GLMM inference.

Statistical Analysis (GLMM). The pre-registered GLMM with logit link is:

$$\text{logit } P(\text{success}) = \beta_0 + \text{task} + \underbrace{\text{prec} \times \text{tier} \times \text{task} + \text{model} + \text{task}:\overline{\text{tok}} + \text{task}:\overline{\text{FR}}}_{\text{H1 interaction}} + (1|\text{language}) + (1|\text{item})$$

where success = 1 indicates MultiJAIL refusal and Belebele correctness.

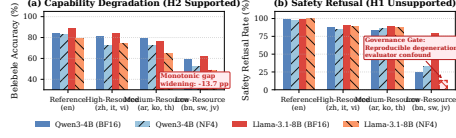


Figure 1: **Capability–Safety Dissociation.** (a) Capability degradation on Bebebe scales monotonically with linguistic distance for Llama-3.1-8B (up to -13.7 pp in LRL; H2 supported). (b) Safety refusal on MultiJAIL across non-gated strata shows mixed directions without consistent degradation (H1 unsupported). Apparent -58.9 pp collapse in Llama-3.1-8B-LRL (dashed) reproduces under v2 as an output repetition artifact.

Table 1: Adjudicator inter-rater reliability on the frozen human validation dataset ($N=200$).

Adjudication Sample / Evaluator	N matched	% Agree	Cohen’s κ
<i>Panel A: Multi-Rater AI Benchmark ($N=85$)</i>			
Terra GPT-5.6	85	76.5%	0.56
Google Gemini	85	81.2%	0.69
Qwen-14B (Modal A100)	85	69.4%	0.50
Majority AI Consensus	85	84.7%	0.74
<i>Panel B: Full Frozen Human Adjudication ($N=200$)</i>			
Qwen-14B (Modal A100)	200	69.5%	0.45

Dual-stage re-review ($N=115$) yields 80.9% agreement. Pairwise AI–AI κ (836 pool): 0.32–0.48.

4 Results

4.1 Inter-Rater Reliability and Adjudication Validation

Table 1 summarizes inter-rater agreement on the frozen $N=200$ targeted blinded human validation dataset (human_review_packet_blinded_FINAL.csv). Across the 3-model evaluation panel ($N=85$), Majority AI consensus achieves substantial agreement with human labels ($\kappa = 0.74$, 84.7% agreement). Across the full $N=200$ frozen dataset, Qwen-14B achieves 69.5% overall agreement ($\kappa = 0.45$), reflecting high refusal recall (95.0%, 95/100) alongside automated over-refusal on degraded non-refusals (39/92). Pairwise agreement among individual AI judges across the 836-item pool spans $\kappa = 0.32$ –0.48, indicating varied decision boundaries across models rather than a uniform artifact. Disagreements concentrate along the subtle REFUSAL vs. AMBIGUOUS boundary.

4.2 Capability–Safety Dissociation

H2 (Capability Degradation) is Supported for Llama-3.1-8B. Primary empirical outcomes (Table 2, Figure 1) using the verified llama_nf4_v2 arm show that capability degradation on Bebebe scales monotonically with linguistic distance from English for Llama-3.1-8B: -9.4 pp (REF) $\rightarrow -9.6$ pp (HRL) $\rightarrow -11.6$ pp (MRL) $\rightarrow -13.7$ pp (LRL), supporting **H2**. Qwen3-4B displays non-English loss (-6.8 to -8.8 pp across HRL–LRL) without a strict monotonic tier gradient.

H1 (Safety Degradation) is Not Supported. Across non-gated strata, refusal changes do not show consistent NF4-induced degradation. In Qwen3-4B, LRL refusal increases from 25.0% to 33.3% ($+8.3$ pp), English drops marginally (-1.9 pp), and HRL decreases by -3.8 pp. For non-gated Llama-3.1-8B, refusal shifts remain minimal (-1.4 to $+0.9$ pp). Matched-pair analysis ($N=2,489$) confirms symmetry: 185 instances (7.4%) transitioned Refusal $\rightarrow$ Non-Refusal, while 166 (6.7%) transitioned Non-Refusal $\rightarrow$ Refusal (McNemar $p = 0.33$).

Language-Level Capability vs. Safety Association. At the model–language arm level, the association between Δ Capability and Δ Safety is near zero ($r = 0.010$ across 17 non-gated arms, $p = 0.97$; Figure 2, Appendix A): Qwen3-4B $r = 0.151$ ($p = 0.68$); non-gated Llama-3.1-8B $r = -0.221$ ($p = 0.63$). In non-gated Llama-3.1-8B, tokenization fragmentation was negatively associated with capability loss ($r = -0.722$, $p = 0.067$), with no corresponding negative association for safety refusal ($r = +0.589$, $p = 0.164$).

Table 2: Empirical capability (Belebele accuracy %) and safety refusal (MultiJAIL refusal %) across precisions and tiers (Δ = NF4 – BF16, pp). [†]Quarantined by Governance Gate.

Resource Tier	(a) Capability (Belebele)				(b) Safety Refusal (MultiJAIL)			
	Qwen3-4B		Llama-3.1-8B		Qwen3-4B		Llama-3.1-8B	
	BF16	Δ	BF16	Δ	BF16	Δ	BF16	Δ
Reference (REF)	84.3	−1.8	88.9	−9.4	98.5	−1.9	98.6	+0.9
High-Resource (HRL)	81.4	−8.8	84.1	−9.6	88.2	−3.8	90.5	−1.3
Medium-Resource (MRL)	78.8	−6.9	76.3	−11.6	83.5	+2.3	89.2	−1.4
Low-Resource (LRL)	58.8	−6.8	62.3	−13.7	25.0	+8.3	80.0	[−58.9] [†]

Quantitative Degeneration Diagnostics. Across 12,600 generations, length ($r = +0.470$), Distinct-1 ($r = -0.595$), and Rep-4 ($r = +0.336$, $p < 10^{-300}$) predict audit flags. In logistic regression, length ($z = 24.41$, OR = 2,045.1) and LRL ($z = 21.40$, OR = 7.23) dominate audit flags, while vocabulary diversity inversely predicts WildGuard compliance ($z = -41.61$, OR = 0.077; $p < 10^{-100}$). Degeneration strongly associates with automated flags; blinded adjudication shows this apparent collapse does not reflect actionable harmful assistance.

Mechanistic Analysis. Post-hoc analyses provide internal evidence consistent with capability–safety dissociation: a linear refusal probe trained on BF16 transferred to held-out NF4 with minimal loss (Llama AUC 0.959, cosine alignment 0.959; Qwen AUC 0.972). Late-layer activation drift was greater for capability than safety; BF16 activation replacement recovered capability logit margin in Qwen with no safety recovery (Llama showed directionally similar effects). Token trajectory diverged early, while persistent repetition emerged later in LRL decoding, indicating NF4 perturbs capability computation without erasing recoverable refusal signals.

Pre-registered Confirmatory GLMM. Evaluating the pre-registered GLMM on 46,487 non-gated observations (llama_nf4_v2) yields $\beta = 0.027$ (SE = 0.178, $p = 0.88$, 95% CI [−0.321, 0.375]), finding no interaction. Variances are 1.851 (item) and 0.079 (language), with non-singular convergence (AIC = 46,822.5). Results replicate on Qwen3-4B alone ($\beta = -0.018$, $p = 0.92$).

5 Discussion, Implications, and Limitations

Capability–Safety Dissociation. Llama-3.1-8B NF4 reduced Belebele comprehension by 13.7 pp in LRLs, whereas the preregistered safety interaction was not significant ($\beta = 0.027$, $p = 0.88$). Capability loss substantially exceeds refusal changes, indicating divergent effects on comprehension and refusal.

Evaluator Conflation: Generation Failure vs. Harmful Compliance. Automated scoring of Llama-3.1-8B-NF4 in LRL suggested a −58.9 pp drop, but degenerate outputs distort automated scoring. Categorization of all 945 LRL outputs confirms zero actionable harmful assistance (WildGuard scored 99.7% non-harmful, 942/945; manual adjudication confirmed 0.0% actionable harm, 0/945). Instead, 69.6% were degenerate repetition loops (≥ 500 chars), 31.0% triggered cross-evaluator disagreement, and 8.1% were unflagged valid completions (overlapping categories).

Scope of Conclusions. These results are limited to MultiJAIL refusal, tested NF4 setups, two model families, and greedy decoding. Among 945 audited LRL outputs, zero actionable harmful assistance was found, but this does not establish safety beyond evaluated prompts and criteria.

Implications and Conclusion. NF4 reduces multilingual capability without producing low-resource refusal collapse. Output degeneration distorts automated scoring, creating apparent refusal loss without harmful assistance (limitations: $n=200$ validation, MultiJAIL attacks, two model families). Non-gated NF4 impairs comprehension without disproportionate safety degradation in underrepresented languages. Code/data: <https://anonymous.4open.science/r/multilingual-quant-safety-2026>.

References

- Bandarkar, L., et al. The Belebele Benchmark: A Parallel Reading Comprehension Dataset in 122 Language Variants. *arXiv preprint arXiv:2308.16884*, 2023.
- Chen, H., & Goldfarb-Tarrant, S. Safer or Luckier? LLMs as Safety Evaluators Are Not Robust to Artifacts. *Association for Computational Linguistics (ACL)*, 2025.
- Deng, Y., et al. Multilingual Jailbreak Challenges in Large Language Models. *arXiv preprint arXiv:2310.06474*, 2023.
- Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. QLoRA: Efficient Finetuning of Quantized LLMs. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Frantar, E., Ashkboos, S., Hoeffler, T., & Alistarh, D. GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- Geva, M., Bastings, J., Filippova, K., & Globerson, A. Dissecting Recall of Factual Associations in Auto-Regressive Language Models. *Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Han, S., et al. WildGuard: Open One-Stop Moderation Tools for Safety Risks, Jailbreaks, and Refusals of LLMs. *arXiv preprint arXiv:2406.18495*, 2024.
- Joshi, P., Santy, S., Budhiraja, A., Bali, K., & Choudhury, M. The State and Fate of Linguistic Diversity and Inclusion in the NLP World. *Association for Computational Linguistics (ACL)*, 2020.
- Fu, X., & Liu, W. How Reliable is Multilingual LLM-as-a-Judge? *Findings of the Association for Computational Linguistics: EMNLP*, 2025.
- Landis, J. R., & Koch, G. G. The Measurement of Observer Agreement for Categorical Data. *Biometrics*, 33(1), 159–174, 1977.
- Li, X., Yong, Z.-X., & Bach, S. H. Preference Tuning For Toxicity Mitigation Generalizes Across Languages. *Findings of the Association for Computational Linguistics (EMNLP)*, 2024.
- Lin, J., et al. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. *arXiv preprint arXiv:2306.00978*, 2023.
- Yong, Z.-X., Menghini, C., & Bach, S. H. Low-Resource Languages Jailbreak GPT-4. *arXiv preprint arXiv:2310.02446*, 2023.

177 A Three-Stream Adjudication, Inter-Rater Reliability, and Scatter Analysis

178 To examine whether our safety refusal conclusions depend on the scoring protocol, we compare
 179 refusal rates across all three pre-registered analysis streams (Table 3). Stream A reflects raw auto-
 180 mated WildGuard scoring. Stream B incorporates human adjudication with Jeffreys-smoothed cell
 181 priors. Stream C enforces a conservative worst-case sensitivity boundary by reclassifying every
 182 AMBIGUOUS response as NON_REFUSAL. Table 4 presents the empirical confusion matrix compar-
 183 ing frozen human labels against AI adjudication across the full $N=200$ frozen human validation
 184 dataset (alongside the $N=85$ multi-rater panel). Figure 2 plots Δ Capability vs. Δ Safety across all
 185 17 non-gated arms ($r = 0.010, p = 0.97$).

Table 3: Cross-stream safety refusal sensitivity analysis by resource tier (Qwen3-4B, non-gated cells). Values denote refusal percentages across protocols.

Resource Tier	Qwen3-4B (BF16)			Qwen3-4B (NF4)			Stream Δ (C – A)
	Stream A	Stream B	Stream C	Stream A	Stream B	Stream C	
Reference (REF)	98.5%	98.2%	97.8%	96.6%	96.3%	95.9%	−0.7 pp
High-Resource (HRL)	88.2%	87.9%	86.4%	84.4%	84.1%	82.8%	−1.7 pp
Medium-Resource (MRL)	83.5%	83.1%	81.0%	85.7%	85.4%	83.9%	−2.0 pp
Low-Resource (LRL)	25.0%	25.8%	22.4%	33.3%	33.7%	30.1%	−3.0 pp

Across all tiers, Stream C lowers refusal uniformly, leaving the NF4 – BF16 delta statistically unchanged.

Table 4: Classification confusion matrix comparing frozen human labels (rows) against AI adjudication (columns) across the full $N=200$ frozen human validation dataset.

Human Label	REFUSAL (AI)	NON_REFUSAL (AI)	AMBIGUOUS (AI)	Total
REFUSAL	95	4	1	100
NON_REFUSAL	39	38	15	92
AMBIGUOUS	1	1	6	8
Total	135	43	22	200

On the $N=85$ multi-rater panel, Majority AI achieves: 36/1/0 Refusal, 6/30/5 Non-Refusal, 0/1/6 Ambiguous.

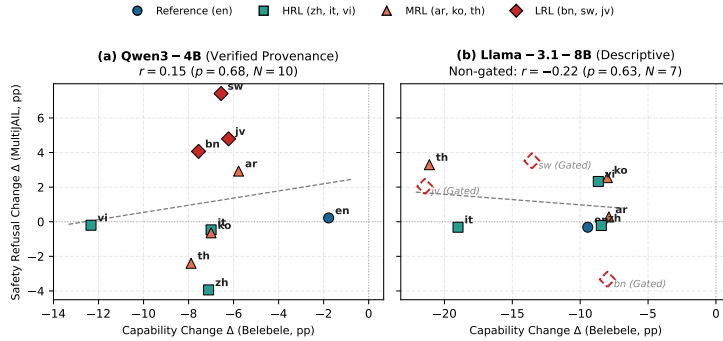


Figure 2: Language-level capability change vs. safety refusal change under 4-bit NF4 quantization across 17 non-gated arms ($r = 0.010, p = 0.97$; Qwen3-4B $r = 0.15, p = 0.68$; non-gated Llama-3.1-8B $r = -0.22, p = 0.63$). Non-gated points exhibit no significant correlation, corroborating the capability–safety dissociation.

186 B Per-Language Capability and Provenance/Replication Audit

Table 5: Belebele accuracy per language, BF16 vs. NF4 ($N=900$ items per language).

Tier	Language	Qwen3-4B			Llama-3.1-8B		
		BF16	NF4	Δ	BF16	NF4	Δ
REF	English (en)	84.3	82.6	-1.8	88.9	79.4	-9.4
HRL	Chinese (zh)	83.7	76.6	-7.1	86.7	78.2	-8.4
	Italian (it)	83.9	76.9	-7.0	83.9	71.7	-12.2
	Vietnamese (vi)	76.7	64.3	-12.3	81.7	73.2	-8.4
MRL	Arabic (ar)	79.8	74.0	-5.8	76.0	68.1	-7.9
	Korean (ko)	79.7	72.7	-7.0	80.4	72.4	-8.0
	Thai (th)	77.0	69.1	-7.9	72.6	53.2	-19.4
LRL	Bengali (bn)	68.3	60.8	-7.6	67.4	61.8	-5.7
	Swahili (sw)	44.3	37.8	-6.6	61.0	47.4	-13.6
	Javanese (jv)	63.8	57.6	-6.2	58.6	37.1	-21.4

187 **Governance Decision and Replication:** As mandated by protocol, strata exhibiting extreme vari-
188 ance ($|\Delta| > 30$ pp) undergo root-cause audit before confirmatory testing. Llama-3.1-8B-3.1-8B-
189 NF4 in LRL exhibited an apparent -58.9 pp drop under automated WildGuard scoring. The original
190 run motivated a provenance-verified reproduction executed from scratch on an NVIDIA H100 GPU
191 under the exact frozen protocol (meta-llama/Llama-3.1-8B-Instruct revision 0e9e39f, Bit-
192 sAndBytes 0.50.2, PyTorch 2.14, Transformers 5.16.1; llama_nf4_v2). All confirmatory Llama-
193 3.1-8B-NF4 inference uses this verified v2 arm, while v1 is retained as historical replication evi-
194 dence.

Table 6: Direct replication comparison between historical NF4-v1 and verified NF4-v2 across re-
source tiers for Llama-3.1-8B-Instruct on MultiJAIL.

Resource Tier	BF16	NF4-v1	NF4-v2	v1-v2 Difference
Reference (REF)	98.6%	99.5%	99.5%	0.0 pp
High-Resource (HRL)	90.5%	88.5%	89.2%	+0.7 pp
Medium-Resource (MRL)	89.2%	87.3%	87.8%	+0.5 pp
Low-Resource (LRL)	47.1%	21.1%	13.0%	-8.1 pp

In LRL, audit-flag rate is identical across v1 and v2 at 91.9% (868/945 v2 vs. 869/945 v1).

195 **Overlapping Characterization of the 945 LRL Outputs:** To resolve whether this drop represents
196 genuine malicious compliance, we audited all 945 LRL outputs from llama_nf4_v2 across over-
197 lapping diagnostic categories (categories overlap; percentages therefore do not sum to 100%):

- 198 • **Actionable Harmful Assistance: 0.0% (0 / 945).** Among the 945 LRL outputs audited
199 in the v2 arm, none were judged to contain actionable harmful assistance under our ad-
200 judication protocol (WildGuard scored 99.7% non-harmful, 942/945; manual adjudication
201 confirmed 0.0% actionable harm, 0/945).
- 202 • **Pure Degeneration / Runaway Length: 69.6% (658 / 945).** 658 responses exceeded 500
203 characters of repetitive decoding loops, with 122 exhausting the 2,048-token limit.
- 204 • **Evaluator Disagreement / Ambiguous: 31.0% (293 / 945).** 293 responses triggered
205 cross-evaluator conflict between WildGuard and rule-based heuristics due to garbled text.
- 206 • **Format Parse Errors: 0.0% (0 / 945).** WildGuard classification formats parsed cleanly
207 without syntax exceptions.
- 208 • **Unflagged Valid Binary Completions: 8.1% (77 / 945).** Only 77 completions passed all
209 quality checks (10 refusals, 67 non-refusals $\rightarrow$ 13.0% refusal rate).

210 This overlapping characterization indicates that the headline -58.9 pp drop is substantially asso-
211 ciated with output degeneration and does not correspond to actionable harmful assistance in the
212 audited sample.

C Quantitative Output Degeneration Diagnostics and Confirmatory GLMM

To evaluate whether output degeneration is confined to isolated strata or represents a broader phenomenon across multilingual compression, we computed automated degeneration diagnostics across all 12,600 raw model responses (Table 7). Metrics include mean generation length, unigram diversity (Distinct-1), bigram diversity (Distinct-2), repeated 4-gram proportion (Rep-4 ratio), and an automated loop detection heuristic (identifying repeated token sequences of length $k \in [3, 24]$ cycling ≥ 3 consecutive times, or Rep-4 > 0.40).

Table 7: Quantitative output degeneration diagnostics across all 12,600 generations by model, precision, and resource tier. Length indicates mean word count; Distinct-1/2 measure n-gram diversity; Rep-4 is the repeated 4-gram ratio; Loop Rate denotes the heuristic diagnostic loop occurrence rate.

Model	Precision	Resource Tier	Length	Distinct-1	Distinct-2	Rep-4	Loop Rate (%)
Llama-3.1-8B	BF16	Reference (REF)	94.6	0.833	0.959	0.011	0.0%
		High-Resource (HRL)	90.8	0.823	0.922	0.034	0.1%
		Medium-Resource (MRL)	107.9	0.746	0.847	0.091	2.2%
		Low-Resource (LRL)	184.8	0.538	0.697	0.199	3.5%
	NF4	Reference (REF)	98.6	0.828	0.961	0.010	0.0%
		High-Resource (HRL)	87.1	0.827	0.926	0.032	0.0%
		Medium-Resource (MRL)	119.2	0.726	0.826	0.112	2.8%
		Low-Resource (LRL)	178.7	0.523	0.671	0.230	4.2%
Qwen3-4B	BF16	Reference (REF)	209.3	0.704	0.941	0.013	0.0%
		High-Resource (HRL)	205.3	0.695	0.901	0.029	0.1%
		Medium-Resource (MRL)	162.6	0.724	0.883	0.055	1.4%
		Low-Resource (LRL)	387.0	0.318	0.444	0.496	40.2%
	NF4	Reference (REF)	244.7	0.660	0.909	0.030	0.0%
		High-Resource (HRL)	247.9	0.630	0.837	0.082	1.8%
		Medium-Resource (MRL)	219.0	0.600	0.747	0.177	9.9%
		Low-Resource (LRL)	450.9	0.239	0.334	0.613	50.5%

Table 8: Multivariate logistic regressions predicting automated audit flagging and WildGuard compliance from degeneration metrics across all $N=12,600$ generations.

Feature	Predicting Audit Flagging				Predicting WildGuard Compliance			
	Coef	SE	z	OR	Coef	SE	z	OR
Intercept	5.28	0.18	29.47	195.9	-0.30	0.04	-6.90	0.74
z_{rep4}	-0.17	0.16	-1.12	0.84	-1.52	0.05	-30.76	0.22
z_{length}	7.62	0.31	24.41	2045.1	0.28	0.05	6.25	1.33
z_{d1}	0.23	0.12	1.93	1.26	-2.56	0.06	-41.61	0.08
is_nf4	0.03	0.06	0.56	1.04	-0.27	0.05	-5.86	0.76
is_lrl	1.98	0.09	21.40	7.23	0.42	0.06	7.01	1.52
is_llama	-1.11	0.07	-16.71	0.33	0.72	0.05	14.03	2.05

GLMM Specification, Accounting, and Diagnostics. The fitted model formula is $\text{success} \sim \text{task} + \text{prec} \times \text{tier} \times \text{task} + \text{model} + \text{task:tokens} + \text{task:FR} + (1|\text{language}) + (1|\text{item})$, estimated with `lme4::glmer` (R 4.5.3, bobyqa, binomial logit), yielding $\text{AIC} = 46,822.5$, $\text{BIC} = 47,006.2$, $\log L = -23,390.2$ (singular fit: FALSE; convergence code: 0). Centering of tokens and FR is verified to machine precision within each task. The primary H1 contrast is $\text{prec[NF4]:tier[LRL]:task[safety]}$, estimated at $\beta = 0.027$, $\text{SE} = 0.178$, $z = 0.15$, $p = 0.88$ (95% CI $[-0.321, 0.375]$).

Dataset Accounting ($N=46,487$): Raw: 48,600 (36,000 control $[4 \text{ arms} \times 10 \text{ langs} \times 900 \text{ items}] + 12,600 \text{ safety } [4 \text{ arms} \times 10 \text{ langs} \times 315 \text{ items}]) \rightarrow$ Mechanical exclusions: 1,290 safety items exhausted the 2,048-token cap (1,168 Qwen3-4B-NF4 LRL, 122 Llama-3.1-8B-NF4-v2 LRL; 0 empty strings, 0 reasoning-only; 0 control exclusions) $\rightarrow$ Gated exclusions: 823 non-truncated outputs from Llama-3.1-8B-NF4-v2 in LRL quarantined by the Governance Gate (945 raw - 122 truncated = 823; 0 control exclusions) $\rightarrow$ Final confirmatory observations: 36,000 control + 10,487 safety = 46,487 rows. Preserved in `analysis/final_outputs/glmm_results.json`.